CoEvoMed: Co-Evolving Supervision Principles and Medical Vision–Language Models

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Abstract

Training medical vision–language models requires supervision that adapts as the learner's capabilities change across tasks. Existing pipelines typically rely on repeated manual error analysis and task-specific training data construction, making this adaptation costly. We introduce CoEvoMed, a co-evolutionary framework that couples a medical vision–language learner with a multi-agent supervision process integrating the evolution of construction principles with curriculum allocation. These reusable principles define question intent, guide how responses draw on medical evidence, and clarify conceptual distinctions in training examples. A revision agent updates these principles in response to the learner's evolving task-specific capabilities, guided by comparisons of factual consistency and task alignment between examples generated from the same medical evidence. Under fixed budgets for each task, curriculum allocation adjusts construction effort according to the learner's capability profile and learning progress. Using CoEvoMed, we train an 8B model supporting both 2D and 3D medical tasks. It surpasses Hulu-Med-7B by 1.15–4.80 percentage points on four 2D VQA benchmarks and achieves higher RaTEScore on MIMIC-CXR and IU-Xray, while delivering strong performance on 3D recognition, question answering, and report generation. Results at 32B further support the scalability of the framework.

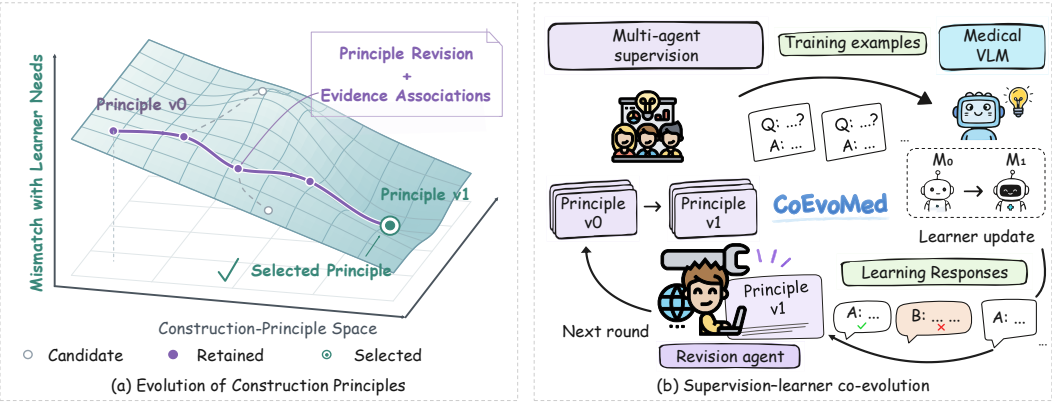


Figure 1: Co-evolution in CoEvoMed. (a) Learning responses guide principle evolution. (b) The multi-agent process connects supervision construction, learner updates, and principle revision.

1 Introduction

The knowledge covered by training data and the organization of its supervision constrain capability formation in medical multimodal models. The former defines the medical evidence available for learning; the latter organizes the task objectives and reasoning relations to be learned from that evidence. Supervisory utility depends on alignment with the learner's current state. As training proceeds, task-specific mastery and learning gains evolve unevenly, limiting the continued suitability of a fixed organization of training supervision. This motivates treating data construction as an endogenous component of training, with supervision construction and training allocation continually adapting to learner development.

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Existing data refinement practices often rely on manual error inspection and failure mode analysis (Wu et al., 2019), followed by targeted supervision construction (Khan et al., 2025). Repeated fine-grained diagnosis and supervision design incur substantial human effort (Ribeiro & Lundberg, 2022) and limit the timely translation of evolving learning needs into systematic training interventions. This limitation motivates a shift from static supervision provision to supervision construction driven by learning responses: observed learning responses inform revisions to reusable construction principles, while realized learning gains regulate the allocation of training resources, establishing a reciprocal evolutionary process between supervision construction and model learning.

Curriculum learning adapts content selection and allocation (Bengio et al., 2009; Matiisen et al., 2017); DataEnvGym uses student feedback for iterative data generation (Khan et al., 2025); Evo-PI evolves principles for evaluating and rewarding medical reasoning (Zheng et al., 2026). We focus on the reciprocal development of construction principles and learner capabilities: learning responses revise how evidence is expressed as supervision, and those revisions shape subsequent learning. This connects the evolution of supervision construction with the allocation of finite construction resources.

We introduce *CoEvoMed*, a framework for co-evolving supervision agents and multimodal learners through *supervision construction driven by learning responses*. CoEvoMed constructs training supervision from medical evidence through reusable principles. The learner updates its parameters using this supervision, while task-specific assessments of its capability profile guide the evolution of the principles and the adaptation of subsequent supervision. Residual errors guide their revision, while realized learning gains regulate construction effort across capabilities under a fixed budget. Multi-agent collaboration compares candidate principles under matched evidence and objectives, evaluating supervisory alignment and factual consistency. Selected principles shape subsequent supervision; the learner's ensuing responses guide further revision. Independent validation sets organized by task supply these responses, a shared support set grounds construction in medical evidence, and independent external benchmarks measure outcomes. The evolution of construction principles and the development of learner capabilities jointly shape the learning trajectory.

Overall, our main contributions are summarized as follows:

- We propose a co-evolutionary framework that incorporates the evolution of construction principles into learner training, coupling supervision construction and capability development through reciprocal feedback.
- We develop a curriculum allocation policy that balances the development of weaker capabilities with the retention of acquired capabilities. Learning progress informs how a fixed construction budget is redistributed as the learner develops.
- We design a multi-agent system that integrates curriculum allocation and principle evolution. Allocation directs construction effort across capabilities, while evolving principles guide how medical evidence is organized into training examples.

2 Related Work

2.1 Curriculum Learning and Adaptive Supervision

Curriculum learning orders examples by difficulty (Bengio et al., 2009); adaptive teaching selects them according to learning progress or learner state (Matiisen et al., 2017; Liu et al., 2017). Beyond selection, Generative Teaching Networks optimize synthetic supervision through meta-gradients (Such et al., 2020), and DataEnvGym guides data generation with learner feedback (Khan et al., 2025). Recent approaches couple task generation, visual supervision, or task knowledge with learner development (Huang et al., 2026a; Zhang et al., 2026a; Wu et al., 2026). CoEvoMed jointly adapts supervision construction and curriculum allocation: reusable construction principles evolve with learning responses, while realized gains guide construction effort across capabilities.

2.2 Multi-agent adaptation and co-evolution

Self-Instruct generates instruction-following supervision (Wang et al., 2023). AutoGen organizes agent conversations (Wu et al., 2024), Graph-of-Agents adapts collaboration structures (Yun et al., 2026), and Automated Design of Agentic Systems searches agent programs through a meta agent (Hu

et al., 2025). Feedback supports output refinement in Self-Refine (Madaan et al., 2023), cross-trial reflection in Reflexion (Shinn et al., 2023), context refinement in ACE (Zhang et al., 2026b), and policy optimization through interaction rewards in CoMAS (Xue et al., 2026). CoEvoMed connects the revision of construction principles with the development of learner capabilities: learning responses revise evidence-grounded construction principles, whose induced supervision shapes subsequent learning.

2.3 Medical multimodal learning

Medical vision-language tasks span radiology, pathology, literature-derived VQA, and report generation (Lau et al., 2018; Liu et al., 2021; He et al., 2020; Zhang et al., 2024; Johnson et al., 2019; Demner-Fushman et al., 2016). LLaVA-Med and Med-Flamingo study biomedical instruction tuning and few-shot adaptation (Li et al., 2023; Moor et al., 2023). RadFM integrates 2D/3D radiological inputs with text (Wu et al., 2025); M3D develops volumetric image–text and instruction data (Bai et al., 2024). MedVLSynther combines source-conditioned generation with rubric-based verification (Huang et al., 2026b), and MMedAgent-RL introduces specialist-reliability curricula (Xia et al., 2026). CoEvoMed studies supervision construction adapted to the learner's capabilities across visual settings and backbone families.

3 Problem formulation

We formalize the co-evolution of construction principles and learner capabilities, with curriculum allocation modeled as a sequential decision problem. At round e, capability-specific shares $\boldsymbol{q}_e$ allocate a finite construction budget across capability cells defined within each task, subject to fixed task-level quotas. Construction principles Σ_e are reusable specifications that determine how medical evidence is organized into supervision for a target capability. These principles define question intent, guide how responses draw on medical evidence, and clarify conceptual distinctions.

3.1 Learner state and curriculum allocation

At round e, the learner M_e is assessed on independent validation sets $\mathcal{V} = \{\mathcal{V}_t\}_{t\in\mathcal{T}}$ for tasks $\mathcal{T}$. Support set $\mathcal{S}$ supplies medical visual evidence and associated text for supervision construction; independent external benchmarks evaluate generalization. Predefined capability cells form $\mathcal{C}$, with each cell c assessed on a fixed nonempty subset $\mathcal{V}_c \subseteq \mathcal{V}_t$. A capability evaluation functional $\mathcal{J}_c$ maps predictive behavior to empirical mastery:

$$\mathcal{J}_c[M_e] = \frac{1}{|\mathcal{V}_c|}\sum_{z\in\mathcal{V}_c} u_c(M_e, z), \qquad \widehat{\boldsymbol{\kappa}}_e = \big(\mathcal{J}_c[M_e]\big)_{c\in\mathcal{C}}. \tag{1}$$

Here u_c is a fixed evaluation rule that assigns a score to each example, with higher scores indicating better performance. The vector $\widehat{\boldsymbol{\kappa}}_e$ is the observable capability profile. Task performance and error evidence constitute observation o_e.

The curriculum planner selects action a_e from history summary $\mathbf{s}_e = \Phi(h_e)$, where h_e records observations, actions, and rewards through round e. We model this interaction as a finite-horizon partially observable Markov decision process (Kaelbling et al., 1998; Matiisen et al., 2017). The complete state X_e includes learner parameters and optimization state, active construction principles, revision history, scheduling information, and remaining resources. Under the rule $\mathscr{E}$ governing principle evolution,

$$P(X_{e+1} \mid X_{0:e}, a_{0:e}) = P_{\mathscr{E}}(X_{e+1} \mid X_e, a_e). \tag{2}$$

The transition comprises supervision construction, verification, learner updates, and principle revision driven by learning responses. Appendix F specifies the state and observation model.

3.2 Supervision construction

Candidate training examples $z = (v, x, y)$ comprise a visual input, instruction, and target response, with distribution

$$\widetilde{p}_e(z) = \sum_{c\in\mathcal{C}} q_{e,c}\, g_{e,c}(z; \Sigma_e, \mathcal{S}). \tag{3}$$

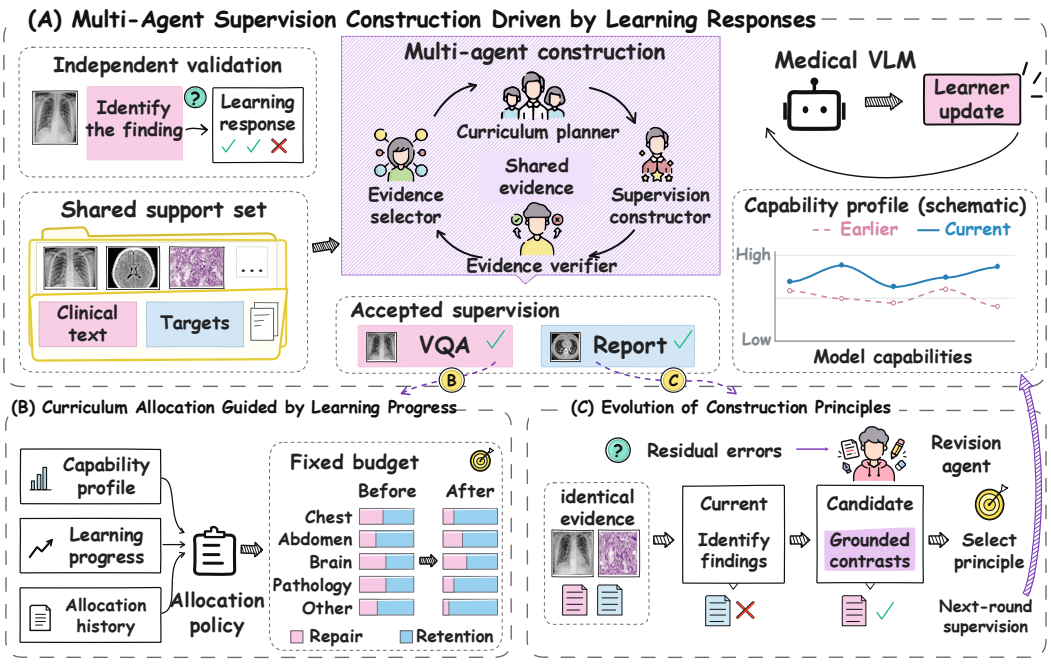


Figure 2: CoEvoMed framework. (A) Multi-agent collaboration constructs and verifies supervision from medical evidence; learner updates yield responses on independent task validation sets. (B) Learning progress guides curriculum allocation under fixed task budgets. (C) The revision agent uses learner responses to revise construction principles and compares supervision generated by current and candidate principles from the same evidence; accepted revisions guide subsequent construction.

Here $g_{e,c}$ denotes the probability distribution over candidate training examples for capability cell c at round e, induced by evidence selection and supervision construction under Σ_e using $\mathcal{S}$, before verification. Nonnegative shares $q_{e,c}$ sum to one. Principle evolution changes the component content, while allocation changes its relative investment. Evidence-grounded verification yields accepted supervision $\mathcal{D}_e^{\text{syn}}$; subsequent mixing determines training exposure.

3.3 Learner Updates and Policy Objective

The training mixture $\mathcal{D}_e^{\text{mix}}$ combines accepted supervision $\mathcal{D}_e^{\text{syn}}$ with permitted original or replay data $\mathcal{R}_e$ reserved for training. The learner update is

$$M_{e+1} = \mathcal{U}(M_e, \mathcal{D}_e^{\text{mix}}), \tag{4}$$

where $\mathcal{U}$ follows the prescribed training procedure and optimization state. Successive capability profiles track the learner's development over E rounds. Responses from M_{e+1} enter h_{e+1} and revise Σ_{e+1} through Equation 9, linking principle evolution to subsequent learning.

Within these dynamics, the allocation policy maximizes cumulative validation progress:

$$\max_{\pi \in \Pi_{\text{adm}}} \mathbb{E}_\pi \left[\sum_{e=0}^{E-1} \gamma^e r_e \right], \tag{5}$$

where Π_{adm} contains policies satisfying the construction budget and task quotas, and $\gamma \in [0, 1]$ is the discount factor. The reward r_e weights validation gains for VQA and report generation equally (Section 4.2). Appendix F.4 relates cumulative gains to performance across rounds.

4 CoEvoMed: co-evolving supervision agents and multimodal learners

CoEvoMed couples supervision construction with learner updates through learning responses (Figure 2). Its multi-agent system integrates curriculum allocation and principle evolution: allocation directs construction effort, while revised principles change how the agents organize medical evidence into supervision.

4.1 Multi-Agent Supervision Construction Driven by Learning Responses

The curriculum planner translates capability feedback into allocation shares and construction targets. The evidence selector retrieves compatible visual records and supported facts from $\mathcal{S}$; the supervision constructor applies Σ_e to form question–answer pairs or reports, and the evidence verifier checks factual consistency, visual dependence, validity, and duplication. Accepted supervision $\mathcal{D}_e^{\mathrm{syn}}$ retains evidence and principle provenance for subsequent assessment.

The learner minimizes negative log-likelihood over response tokens in $\mathcal{D}_e^{\mathrm{mix}}$ under the prescribed budget (Appendix B). After each update, CoEvoMed aligns successive predictions on the same task-validation cases. The observed increment for cell c is

$$\widehat{\Delta\kappa}_e(c) = \mathcal{J}_c[M_e] - \mathcal{J}_c[M_{e-1}], \quad e \geq 1. \tag{6}$$

Paired changes distinguish resolved deficiencies, regressions in acquired capabilities, and persistent gaps, identifying repair and retention needs beyond net progress. The summary $\mathbf{s}_e$ combines current mastery, these changes, and allocation history to guide curriculum allocation. Residual errors in the learner's responses reveal task-specific gaps between current capabilities and task demands; their semantics guide construction targets and principle revision.

An evolution window groups training rounds for assessing principle usage and learning responses. At each window's end, the revision agent proposes candidates from residual errors. Comparisons under matched evidence determine which revisions guide subsequent supervision construction (Section 4.3). The complete procedure appears in Appendix A.

4.2 Curriculum Allocation Guided by Learning Progress

The curriculum planner uses the curriculum allocation policy to select $a_e \sim \mu_{\theta_e}(\cdot \mid \mathbf{s}_e)$, balancing repair and retention and deciding whether to focus on capabilities with the strongest error evidence. Repair targets unresolved gaps in weaker capabilities and recent regressions; retention maintains coverage of acquired capabilities. The allocation map $\mathcal{A}$ translates the action and historical evidence into construction shares:

$$\boldsymbol{q}_e = \mathcal{A}(a_e, h_e), \qquad \sum_{c \in \mathcal{C}_t} q_{e,c} = \bar{q}_t, \tag{7}$$

where $\mathcal{C}_t$ contains the cells of task t and $\bar{q}_t$ is its prescribed share. Thus, adaptation redistributes investment within fixed task budgets. These shares weight candidates in Equation 3; verification and mixing determine realized training exposure.

Let U_t^{val} evaluate the learner on $\mathcal{V}_t$ and $\delta_{e,t} = U_t^{\mathrm{val}}(M_{e+1}) - U_t^{\mathrm{val}}(M_e)$. The reward balances VQA tasks $\mathcal{T}_{\mathrm{V}}$ and report tasks $\mathcal{T}_{\mathrm{R}}$:

$$r_e = G_e = \frac{1}{2|\mathcal{T}_{\mathrm{V}}|} \sum_{t \in \mathcal{T}_{\mathrm{V}}} \delta_{e,t} + \frac{1}{2|\mathcal{T}_{\mathrm{R}}|} \sum_{t \in \mathcal{T}_{\mathrm{R}}} \delta_{e,t}. \tag{8}$$

Evaluation scales remain fixed, so the signed reward retains gains and regressions. Behavior cloning initializes the curriculum planner. PPO with generalized advantage estimation then optimizes Equation 5 through on-policy interactions with the learner (Schulman et al., 2017; 2016). Appendix B specifies the mixed policy and optimization settings.

4.3 Evolution of Construction Principles

Curriculum allocation controls construction effort across capabilities. Construction principles Σ_e specify question intent, evidence–response relations, and conceptual distinctions; their suitability

Table 1: Medical VQA accuracy (%). Bold marks CoEvoMed; its 8B scores use R7.

Models	PMC-VQA	VQA-RAD	SLAKE	PathVQA
Proprietary Models				
Gemini 3 Flash	62.62	69.62	74.55	52.03
GPT-5.6-sol	58.89	67.18	69.15	44.72
Kimi-K3	64.48	73.61	77.13	53.90
MiniMax-M3	57.33	68.07	71.44	57.80
Qwen3.8-Max	64.58	76.81	85.05	64.11
General-purpose Multimodal VLMs				
Qwen3-VL-8B	54.13	56.76	64.84	42.77
Qwen3.5-9B	60.71	68.74	71.87	49.64
Llama3.2-11B	48.10	58.80	65.80	32.90
InternVL3-14B	54.10	66.30	72.80	48.00
Qwen3.6-27B	62.50	71.62	75.07	46.93
Qwen3.8-27B	60.33	68.51	73.93	45.11
Qwen3-VL-32B	57.24	65.85	71.87	44.28
InternVL3-38B	56.60	65.40	72.70	51.00
Medical Multimodal VLMs				
Models $< 10B$				
MedGemma-4B-IT	49.20	72.30	78.20	48.10
HuatuoGPT-V-7B	53.10	67.60	68.10	44.80
Lingshu-7B	56.30	67.90	83.10	61.90
GMAI-VL-7B	56.09	57.21	59.69	39.05
MedVLSynther-7B	55.43	77.57	72.36	65.56
Hulu-Med-7B	66.80	78.00	86.80	65.60
CoEvoMed (8B)	**67.95**	**82.48**	**91.60**	**69.40**
Models $> 10B$				
HuatuoGPT-V-34B	56.60	61.40	69.50	44.40
Lingshu-32B	57.90	76.70	86.70	65.50
MedDr-40B	13.90	65.20	66.40	53.50
Hulu-Med-32B	69.40	81.40	85.70	67.30
CoEvoMed (32B)	**70.20**	**82.71**	**92.60**	**69.60**

depends on the learner's capability state. CoEvoMed revises these reusable principles as learning needs evolve.

At each window's end, sufficiently used principles with nonpositive net repair are considered for revision, where net repair counts repaired cases minus regressed cases. The semantics of residual errors guide principle revision while the target capability and medical facts remain fixed. For example, localization errors motivate more explicit associations between regional evidence and the target response.

Current and candidate principles generate supervision from identical evidence under matched quotas. Selection assesses factual consistency and task alignment, including whether revisions express the targeted evidence associations, conceptual contrasts, or fact coverage. Usable yield (the proportion of evidence assignments producing usable supervision) and paired coverage (the proportion producing usable supervision under both principles) provide screening criteria; Appendix B specifies their thresholds.

Accepted revisions persist as active construction principles (Zhang et al., 2026b). Their supervision shapes learner updates, whose validation responses guide further revision:

$$\Sigma_{e+1} = \mathscr{E}(\Sigma_e, h_{e+1}, \mathcal{S}). \tag{9}$$

Here h_{e+1} includes observations and rewards after the learner update. The operator retains active principles between evolution boundaries and when candidates are rejected.

5 EXPERIMENTS

5.1 EXPERIMENTAL SETUP

The 2D suite comprises VQA-RAD, SLAKE, PMC-VQA, and PathVQA (Lau et al., 2018; Liu et al., 2021; Zhang et al., 2024; He et al., 2020), plus MIMIC-CXR and IU-Xray report generation (Johnson et al., 2019; Demner-Fushman et al., 2016). 3D evaluation covers closed-ended M3D-VQA, open-ended 3D-RAD, and AMOS-MM reports (Bai et al., 2024; Gai et al., 2025; Ji et al., 2024). Qwen3-VL-8B-Instruct (Bai et al., 2025a) is the primary learner; Gemma-3-4B-IT (Gemma Team, 2025) and InternVL3-9B (Zhu et al., 2025) test cross-backbone applicability, with InternVL results in Appendix C.5. Task validation guides adaptation, support evidence grounds supervision, and external benchmarks measure outcomes. Published baselines include Hulu-Med (Jiang et al., 2025), Lingshu (LASA Team et al., 2025), MedVLSynther-7B (Huang et al., 2026b), and CURE-4B (Messina et al., 2026).

5.2 MEDICAL MULTIMODAL PERFORMANCE

CoEvoMed-8B exceeds Hulu-Med-7B by 1.15–4.80 accuracy points across the four VQA benchmarks, while CoEvoMed-32B leads all compared models on each benchmark (Table 1). Report generation reaches 68.22 on IU-Xray at 8B and 64.33 on MIMIC-CXR at 32B (Figure 3). All six 8B results use R7, selected for the best overall cross-task balance.

From the same Qwen3-VL-8B initialization, CoEvoMed outperforms direct training on approximately 700,000 raw examples across all four VQA benchmarks while using fewer training examples (Appendix D.3). This comparison supports adaptive supervision construction and curriculum allocation as a source of performance gains beyond the volume of raw data.

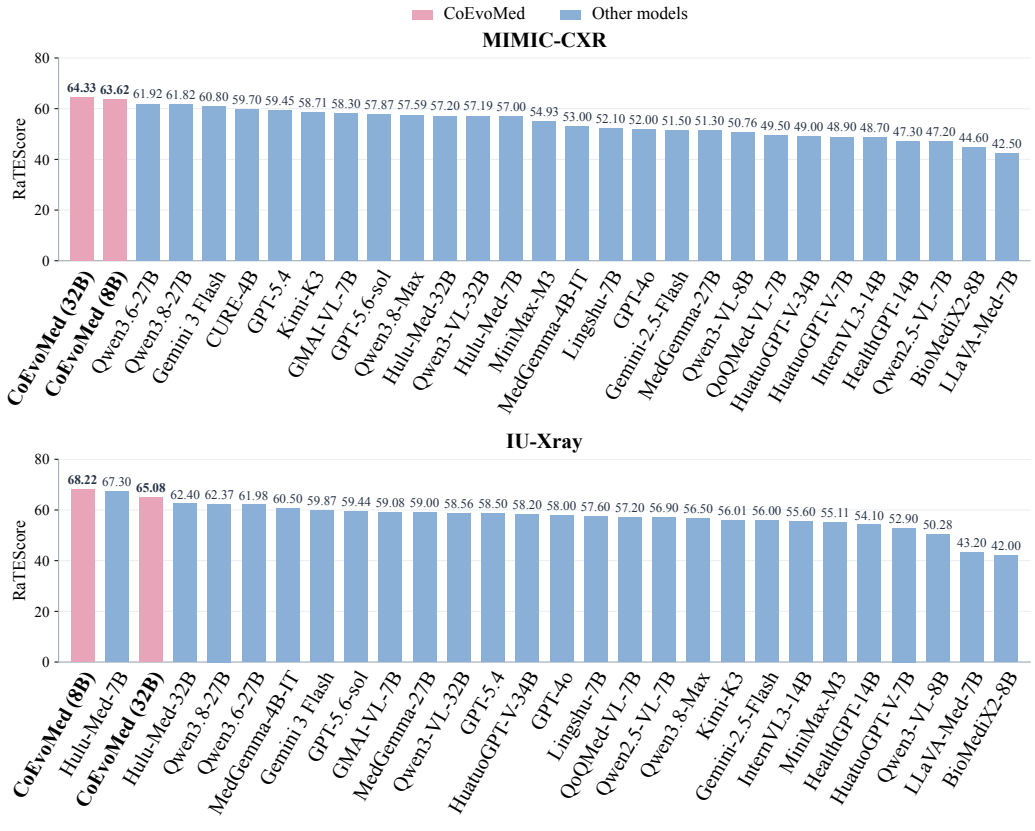


Figure 3: Report generation on MIMIC-CXR (top, 29 models) and IU-Xray (bottom, 28 models), ranked separately by RaTEScore ($\times 100$). Pink denotes CoEvoMed; blue denotes baselines. CURE-4B uses AGRG-29 + GRG. Supplementary metrics appear in Figure 8.

Applicability to volumetric data. CoEvoMed ranks first in the three benchmark summaries in Figure 4: M3D-VQA macro accuracy, 3D-RAD macro ROUGE-1, and AMOS-MM RaTEScore. These results extend the evaluation to volumetric question answering and report generation. Category-level results, supplementary metrics, and evaluation details appear in Appendix C.

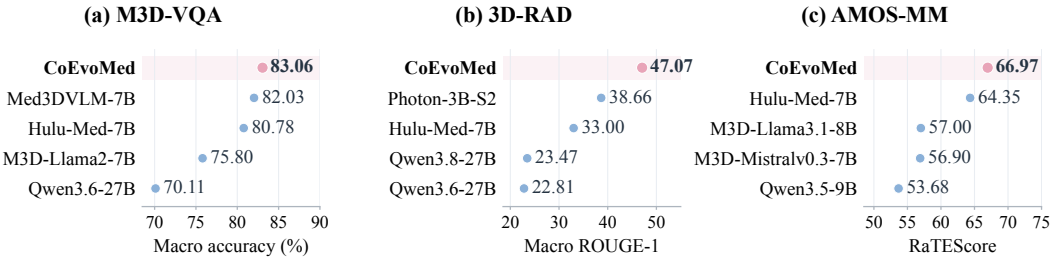


Figure 4: Volumetric evaluation: (a) M3D-VQA macro accuracy; (b) 3D-RAD macro ROUGE-1; (c) AMOS-MM RaTEScore. Scores are scaled to 100. Each panel shows CoEvoMed (pink) and the four highest-scoring baselines (blue) for its metric within the comparison set in Appendix C.

5.3 Learning dynamics and mechanism analysis

We examine learning trajectories, feedback–learner alignment, and stage-dependent principle utility. Mechanism comparisons use VQA-RAD and MIMIC-CXR; full results and experimental details appear in Appendix D.

Feedback–learner alignment. From the same R5 checkpoint, we jointly vary numerical capability statistics and summaries of error semantics using R2, R3, or R5 validation feedback. Current feedback improves VQA-RAD accuracy by 2.62 percentage points and MIMIC-CXR RaTEScore by 2.77 points, outperforming historical feedback (Table 2, panel A). This supports aligning curriculum allocation and supervision construction with current capabilities and error patterns.

Dependence on training stage. Original and evolved principles show stage-dependent utility at checkpoints after rounds 2 and 5 (R2 and R5) (Table 2, panel B). On VQA-RAD, original principles lead by 4.22 points at R2, whereas evolved principles lead by 8.26 at R5. On MIMIC-CXR, their difference grows from 0.12 to 9.21 points in favor of evolved principles. This stage dependence supports revising supervision as the learner develops; all six benchmarks appear in Appendix D.2.

Table 2: Mechanism ablations: (A) the same R5 learner with R2, R3, or current R5 feedback; (B) original versus evolved construction principles at R2 and R5. VQA-RAD reports accuracy (%); MIMIC-CXR reports RaTEScore ($\times 100$). Reference scores are measured before further training.

A. Feedback source at R5

Setting	VQA-RAD	MIMIC-CXR
R5: Reference	73.61	59.00
R2 feedback	72.51	59.11
R3 feedback	71.84	58.16
R5 feedback	76.23	61.77

B. Dependence on training stage

Setting	VQA-RAD	MIMIC-CXR
R2: Reference	61.64	53.31
R2: Original	63.42	56.35
R2: Evolved	59.20	56.47
R5: Reference	73.61	59.00
R5: Original	67.97	52.56
R5: Evolved	76.23	61.77

Cross-backbone applicability. On Gemma-3-4B-IT, R7 attains 70.73% semantic VQA accuracy and 65.89 MIMIC-CXR RaTEScore, improving over Base by 22.39 and 16.23 points (Figure 5). The same checkpoint ties for the highest observed VQA accuracy and achieves the highest report score, supporting applicability beyond Qwen at a smaller model scale.

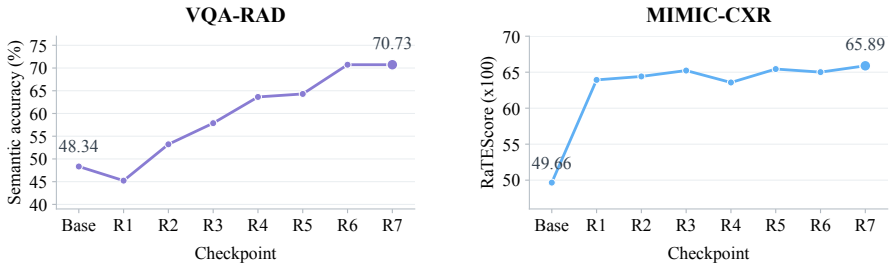


Figure 5: Cross-backbone evaluation on Gemma-3-4B-IT from Base through R7. VQA-RAD uses semantic accuracy (%); MIMIC-CXR uses RaTEScore scaled by 100. Both R7 scores belong to the same checkpoint.

Learning trajectories. Figure 6 shows that attribute queries improve, organ recognition stabilizes, and localization recovers after declining over R1–R6. These divergent trajectories indicate that capability development is not uniform across tasks, providing the state information used for principle revision and construction allocation. Figure 7 shows gains on SLAKE, PathVQA, and IU-Xray, peaking at R7–R8. Their agreement with the task-level profiles connects learner responses to externally measured benchmark performance. Together, the profiles characterize capability development and performance on independent external benchmarks.

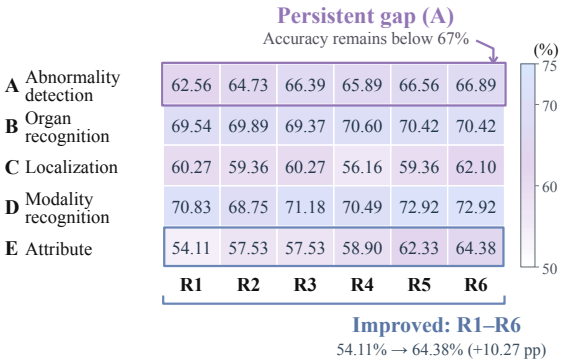


Figure 6: Validation accuracy (%) of the Qwen3-VL-8B learner across five selected QA task categories after completed training rounds R1–R6.

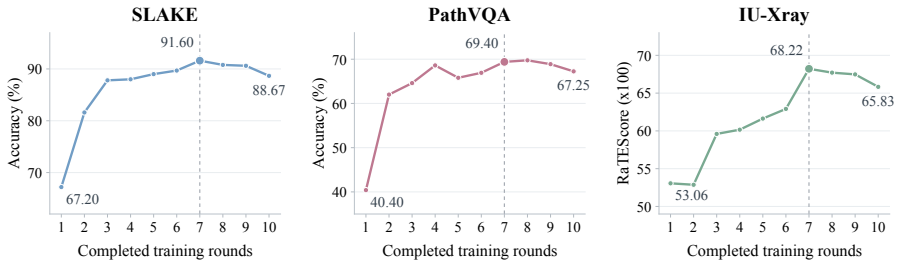


Figure 7: CoEvoMed-8B trajectories. Dashed lines mark R7, used for all six main 2D results.

6 Conclusion

CoEvoMed treats supervision construction as an evolving component of medical vision–language learning. Construction principles develop in response to the learner's task-specific capabilities, while the resulting supervision shapes subsequent capability development. Curriculum allocation complements this process by distributing a fixed construction budget according to observed learning progress. Multi-agent collaboration connects these two forms of adaptation, using medical evidence to produce supervision whose content and allocation evolve together.

Experiments across 2D VQA, report generation, 3D tasks, and different backbones support the effectiveness of this framework. Comparisons across feedback sources and training stages further show the value of matching supervision construction to the learner's current capability profile. These findings support joint adaptation of how supervision is constructed and where construction effort is allocated as a basis for improving medical multimodal learning.

AI USE STATEMENT

We used AI tools to assist with manuscript writing and language editing. We reviewed the text and take responsibility for the final manuscript.

ETHICS STATEMENT

This methodological study uses existing publicly available or authorized research datasets, without new patient recruitment or clinical intervention. Relevant risks include unsupported medical statements, annotation-error propagation, and amplification of source-data or model biases. CoEvoMed incorporates evidence-grounded construction and verification to address factual inconsistency in synthesized supervision. Data access and redistribution of derived material remain subject to the original datasets' licenses, access conditions, and privacy requirements. Clinical deployment would require independent validation, assessment across patient populations, and qualified clinical oversight.

REPRODUCIBILITY STATEMENT

Sections 3–4 specify the formulation and mechanisms; Appendix A provides training pseudocode. Appendix B documents learner training settings, curriculum-planner initialization, PPO hyperparameters, supervision budgets and composition, principle-acceptance criteria, agent models, and hardware. Appendix C describes 2D and volumetric evaluation splits and inputs, dataset sources and links, evidence preparation, task-validation construction, metrics, checkpoint selection, and feedback–evaluation separation. Appendix D gives ablation settings and full results, and Appendix F provides formal definitions and the cumulative-gain derivation.

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